

BMC204:Database Management Systems

THEORY ASSIGNMENT

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UNIT 1

1. What is database management system? Describe the features of a database system and how is it different from file system?
2. Define data models. Compare various types of data models.
3. Discuss the three level (ANSI/SPARC) architecture of DBMS. How Physical data independence is different from Logical data independence.
4. Discuss Primary Key, Candidate Key and Super Key by taking examples.
5. What is ER Diagram? Explain different Components of an ER Diagram with their Notation. Also make an ER Diagram for Draw an ER Diagram of College Mgt. System.
6. Discuss various types of Entity, Attribute and Relation used in designing an ER diagram.
7. What are relational model mapping cardinalities? Give example for each cardinalities used in DBMS.
8. Write short note on:
 - I. Participation constraint in ER diagram with a suitable example.
 - II. Aggregation
 - III. Specialization
 - IV. Generalization
9. In an organization several projects are undertaken. Each project can employ one or more employees' each employee can work on one or more projects. Each project is undertaken on the request of client. A client can request for several projects. Each project has only one client. A project can use a number of items and an item may be used by several projects. Draw an E-R diagram and convert it to a relational schema.
10. Consider the assumptions
 - a) Employee works for a department
 - b) In one department there are many employees
 - c) Each employee has some dependents
 - d) Employee works in some projectsDraw an E-R Diagram. Convert the diagram to Relational Model.

UNIT 2

1. Explain following types of constraints with example: Entity Integrity, Referential Integrity and Domain Constraint?
2. Discuss different operations of relational algebra with examples. How is it different from Relational Calculus?
3. Discuss Primary Key, Unique Key and Foreign Key by taking examples. What will be the possible super keys for the relational schema R(E,F,G,H) with EF as Key.
4. A relation named STUDENT (Roll, Name, DOB, Address, AdharNo, Email) is there. Find out the following:
 - a) Alternative Keys
 - b) Non-key Attributes
 - c) Non-Prime attributes
 - d) Prime Attribute
5. Determine the Relational Algebra expression for following- For a relation Employee (EID, Ename, city), find the employees who either live in Delhi or live in Mumbai.
 1. Use Relational Algebra to express the following queries
Suppliers (Sid: Integer, Sname: String, Address: String)
Parts (Pid: Integer, Pname: String, Colour: String)
Catalog (Sid: Integer, Pid Integer, Cost: Real)
The key fields are underlined.
(a) Find the names of Suppliers who supply some red part.
(b) Find the Sids of Suppliers who supply some red part and Address as, '221 XYZ'.

- (c) Find the Sids of Suppliers who supply every part.
- (d) Find the Sids of Suppliers who supply every red Part and every green part.
- (e) Find the pairs of parts supplied by at least two different suppliers.
- (f) Find the Pids of parts supplied by every supplier at less than Rs. 200.

6. Write short notes on

- a. View
- b. Index
- c. Cursor
- d. Procedure
- e. Function

7. What is trigger and what are the uses of a trigger? How is it different from procedure?

8. Consider the relations:

PROJECT (proj #,proj_name,chief architect)

EMPLOYEE (emp#, emp_name)

ASSIGNED (proj#, emp_name)

Use relational algebra to express the following queries:

- (i) Get details of employees working on project
- (ii) Get the employee number of employees who work on all projects.
- (iii) Get details of project on which employee with name 'RAM' is working.

PRACTICAL based QUESTIONS

1. There is a table named CUSTOMER (ID, Name, DOR). Write the sql query to display all the customer details.(Early Date of registration must be displayed first).
2. Write the SQL command to Create the table names STUDENT

Column Name	Data Type	Size	Remark
Roll	Number	5	Primary Key, must start with 'S'
Name	Varchar2	25	Null values are not allowed
DOB	Date		
Email	Varchar2	30	Must take unique values only
DOR	Date		Must not less than DOB

3. Create a table Employee (Emp_id, Emp_name, Emp_add, Emp_basicpay) and insert some records in it. Create another table Emp(Emp_name, Emp_add) from Employee table by using SELECT statement.
4. Create the tables Supplier (SNo, SName, City), Parts (PNo, Pname, Color, City), and Shipment (SNo, PNo, Quantity). Answer the following queries :

- a. Get the total number of suppliers.
- b. Get the total numbers of suppliers currently supplying parts?
- c. Get part numbers of all parts supplied.
- d. Change the color of parts from red to black.

5. Find out the tables named EMP(Id, Name, DOJ, DeptID, Salary), Dept(DeptId, Name, Location)

- a. Display the employee details along with the department name where the employee is working.
- b. Display the employees according to date of joining. (Recent joins should come first)
- c. Display the number of employees working in MCA
- d. Display the employee names whose name starts with 'Ra'.
- e. Display the employee who are getting more salary than what Rohan is getting.

6. Consider the following relational database, where the primary keys are underlined:

Employee(person_name, street, city)

Works(person_name, company_name, salary)

Company(company_name, city)

Manages(person_name, manager_name)

Write the SQL command to execute the following queries:

- a. Find the name of all employees who live in the same city and on the same street as do their managers.

- b. Find the name of all employees in this database who do not work for "First Bank Corporation".
- c. Find the name of all employees who earn more than every of "Small Bank corporation".
- d. Give all managers in this database a 10 percent salary raise.

7. Consider the following relations:

STUDENT (NAME, ROLL_NUMBER ADDRESS, MAIN)
 ADMISSION (ROLL_NUMBER, COURSE, SEMESTER)
 FACULTY (COURSE, FACULTY, SEMESTER)
 OFFERING (BRANCH, COURSE)

Write the SQL command to find the following information:

- a. The name of students admitted in a particular course in a given semester.
 - b. Students who have taken all courses offered by a given faculty.
8. Consider the following tables or relations: Sales people, (SNUM, SNAME, CITY, Mobile) Customer (CNUM, CNAME, CITY, SNUM), Order (ONUM, Amount, Order-data, CNUM, SNUM)

Write the following queries in SQL.

- a. List the customer name, city who belong to 'Mumbai'
- b. List the Customer number, Customer name and City who have ordered for amount more than 100.
- c. List the entire details of sales people who have supplied the order number greater than 101 and less than 200 on a particular date.
- d. List the entire details of sales people who belong to same city as Customer's city.
- e. Display the entire order number supplied by a salespeople 'XYZ'.

9. Consider the following relational database, where the primary keys are underlined:

SUPPLIER (supplier number, Supplier name)
 PART (part number, part name)
 PROJECT (project number, project_name)
 SUPPLY (supplier number, part number, project number)

Write the SQL command to execute the following queries:

- a. Find the part number that is supplied to exactly two projects.
 - b. Find the name of suppliers who supply more than two parts to project 'J1'.
 - c. Find the part numbers that are supplied by every supplier.
 - d. Find the project names that are supplied by supplier 'S1' only
10. Write a trigger which will insert the updated salary, the old salary of an employee along with the date of updation into a table named Employee_Backup.

UNIT 3

1. Define anomalies in database. How normalization helps in removing anomalies?
2. Define Functional Dependency, Full Functional Dependency, Trivial Dependency, multi valued dependency and Join Dependency with examples.
3. How will you check that the decomposition of a relation on a given set of functional dependencies is dependency preserving? Discuss with the help of an example.
4. What is highest normal form of the Relation R(W,X,Y,Z) with the set F= { $WY \rightarrow XZ$, $X \rightarrow Y$ }
5. Consider a relation R(A,B,C,D,E) with set F= { $A \rightarrow CD$, $C \rightarrow B$, $B \rightarrow AE$ } What are the prime attributes of this Relation and Decompose the given relation in 3NF.
6. Prove that if in a relational schema, the number of attributes in a primary key is one; the schema will be at least in 2NF. Consider the relation R(a,b,c,d) with Set F={ $a \rightarrow c$, $b \rightarrow d$ }. Decompose this relation in 2 NF.
7. Explain the Loss Less Decomposition .Suppose that we decompose the schema R(A,B,C,D,E) into R1 and R2 as R1: {A,B,C} R2: {C,D,E} justify whether it is lossless decomposition or not.
8. Normalize the given relation up to 3NF: R: {A,B,C,D} F={ $AB \rightarrow D$, $AC \rightarrow BD$, $B \rightarrow C$ }
9. Consider a relation R (A, B, C, D, E) with following FD's: $A \rightarrow BC$, $CD \rightarrow E$, $B \rightarrow D$, $E \rightarrow A$. Does $B \rightarrow CD$ Exist over the relation R?
10. What do you mean by lose-less decomposition of a relation? Consider the relational scheme R (A,B,C,D,E,F) and FDs: $AB \rightarrow C$, $C \rightarrow A$, $D \rightarrow E$, $F \rightarrow A$, $E \rightarrow D$. Is the decomposition of R into R1 (A, C, D), R2 (B, C, D) and R3 (E, F, D) lossless?
11. Explain basic inference rules applied on the set of functional dependencies. Suppose R with attributes A, B, C, D, E, F have the FDS: $A \rightarrow BC$, $B \rightarrow E$, $CD \rightarrow EF$. Prove that FD: $AD \rightarrow F$ also holds in R.

12. Given a relation R(P, Q, R, S, T) and Functional Dependency set $FD = \{ PQ \rightarrow R, S \rightarrow T \}$, determine whether the given R is in 2NF? If not convert it into 2 NF.
13. Define minimal cover. Find a Canonical cover for the set A set of FD's for the relation R- {A,B,C,D,E,F} is $AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, EC \rightarrow FA, CF \rightarrow BD, F \rightarrow E$.
14. A set of FD's for the relation R- {A,B,C,D,E,F} is $AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, EC \rightarrow FA, CF \rightarrow BD, F \rightarrow E$: Find a Canonical cover for this set.
15. Suppose a relational schema R (P, Q, R, S), and set of functional dependency as following $F: (P \rightarrow QR, Q \rightarrow R, P \rightarrow Q, PQ \rightarrow R)$. Find the Canonical Cover (Minimal Cover).

Unit 4

1. Describe transaction and its various states.
2. What are ACID properties of Transaction?
3. What is Conflict Serializable Schedule? Check the given Schedule S1 is Conflict Serializable or not?
S1: R1(X), R2(X), R2(Y), W2(Y), R1(Y), W1(X)
4. How serial schedule is different from non-serial schedule? What is a serializable schedule? A schedule is given as follows S: r1(X), 12(X), w1(X), r1(Y), w2(X), W1(Y)
Check whether schedule S is a conflict serializable schedule or not?
5. Explain the method of testing the Serializability. Consider the schedule S1 and S2 given below
S1: R1(A), R2(B), W1(A), W2(B)
S2: R2(B), R1(A), W2(B), W1(A)
Check whether the given schedules are conflict equivalent or not?
6. What do you mean by serial and serializable schedules? Explain View Serializability.
7. What are various reasons for transaction failure? A transaction is failed and enters aborted state. Mention the conditions under which we can restart the transaction and kill the transaction.
8. What is a deadlock? Discuss any one deadlock detection algorithm in database transaction processing.
9. Explain Deadlock Handling with Suitable Example.
10. Write short notes on following
 - (i) Testing of serializability
 - (ii) Deadlock Prevention Protocols
 - (iii) Concurrency control in Distributed System
 - (iv) Deferred and Immediate database modification.
11. What is the system log used for? What are the typical kinds of records in a system log? What are transaction commit points, and why are they important?
12. Explain the check point mechanism used to recover from a transaction failure.
13. What do you mean by deadlock prevention? Discuss the Wait-Die and Wound-Wait deadlock prevention algorithm.

Unit 5

1. What are Concurrent Transactions? What are various problems encountered during concurrency control? Discuss each of them.
2. Discuss different locking techniques for concurrency control?
3. Explain Time Stamp Based and Validation Based Concurrency Control technique.
4. Differentiate between
 - a) Two-phase locking protocol and Strict Two-phase locking protocols.
 - b) Basic timestamp and strict timestamp in concurrency protocol.
5. Explain Recovery from Concurrent Transaction.
6. Describe multi-version concurrency control. What are its benefits and disadvantages in comparison to locking?
7. What is Granularity? Discuss multiple granularity.
8. Demonstrates how Amazon used **Oracle Database** to manage **large-scale enterprise applications**.